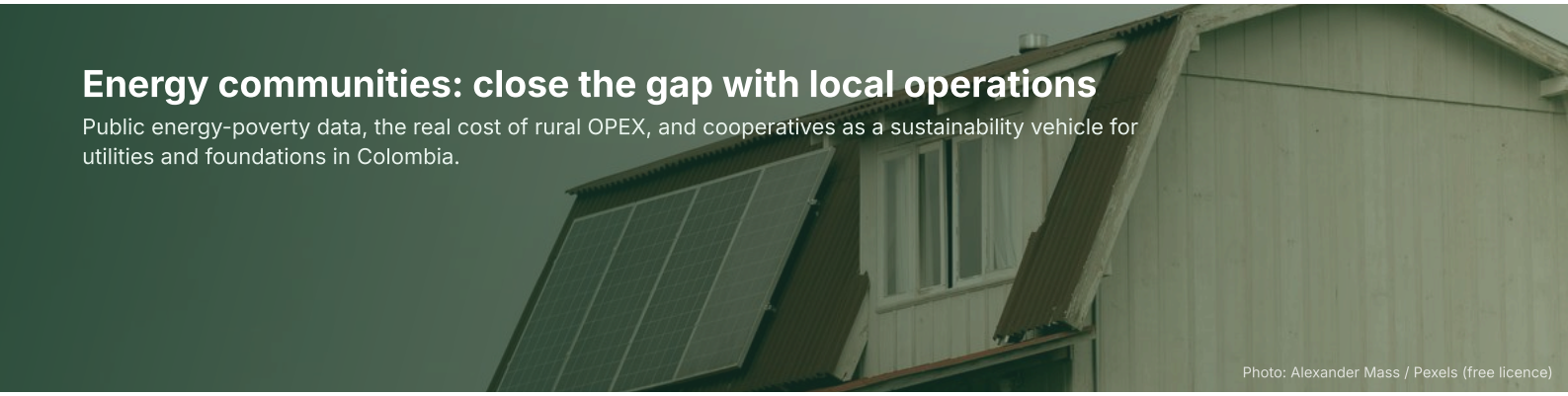


Energy communities: close the gap with local operations

Public energy-poverty data, the real cost of rural OPEX, and cooperatives as a sustainability vehicle for utilities and foundations in Colombia.

Photo: Alexander Mass / Pexels (free licence)

1 The national gap

15.4%
IMPE incidence 2024 (~8.1 M people)[1]

9.5%
2030 scenario target (~5.17 M; −2.9 M)[1]

75.9% / 93.1%
Rural vs national residential coverage (ICEE 2024)[2]

63.1% / 52.8%
IMPE La Guajira / Chocó (2024)[1]

Observed pace ~300 k people/year → 2.9 M gap in ~9.7 years (~2034). 2030 scenario needs ~490 k people/year.[1]

Current pace ~300 k/yr
~2034

2030 scenario ~490 k/yr
2030

2 OPEX, not just CAPEX

Contraloría
No methodology to pay for O&M of microgrids and PV in remote areas.[3]

Amazonas
~$30,000 M (royalties) in PV not operating; damaged batteries — abandonment risk.[4]

CREG
Thousands of rural PV systems with no management → often abandoned.[5]

[1] IMPE Report 2025 (2024 base), pp. 12, 22, 33. [2] UPME ICEE Bulletin 2024. [3] Contraloría, remote solar projects. [4] Seguimiento.co / Procuraduría, Amazonas. [5] CREG GEN-Renewables report; Concept 4022/2019 (O&M).

3 Cooperatives as the tool

CTH PROPOSAL **Community O&M + revenue + local ownership.** OPEX contracts (PV, batteries, monitoring; CREG/SSPD framework) and cooperative governance. KPIs: IMPE access/quality dimensions — not a substitute for ICEE coverage.

4 Ecosystem startups

SC — SunColombia
Bogotá · CleantechHUB portfolio client
Design, build, and O&M for on/off-grid solar for institutions and non-interconnected zones.
898 kWp · 1,122 users (DISPAC, La Guajira)[11]
suncolombia.co

SE — SEOS Energy
Bogotá
Platform linking homes and SMEs to certified installers and solar financing.
75+ SOLvers · 18 departments[12]
seosenergy.co

SL — Soluna Energía
La Guajira · Vichada
Prepaid pay-as-you-go solar for off-grid households in the Caribbean and Orinoquía.
solunaenergia.com

TG — Tierra Grata
Cartagena
Decentralized energy and microfinance in remote rural communities (PayGo model).
1,947 clean energy solutions[13]
tierragrata.org

TW — TuWatt
Bogotá
Structures and operates distributed generation and energy-community projects with traceability.
tuwatt.co

Local talent: Escuela Colombiana de Ingeniería Julio Garavito — 101.76 kWp lab, renewable diplomas, and energy-sector challenges (Solucionatón 10X).[6]

5 PASO Colombia

Productive communities in high energy-poverty departments. **PASO does not run energy projects.**[10]

- Fonseca, La Guajira (IMPE 63.1%)
- Puerto Asís / Guzmán, Putumayo (58.7%)
- Istmina, Chocó (52.8%)
- Arauca (27.9%)

6 Data lab

Funnel from MinEnergía-prioritized communities → shortlist (solar resource, non-interconnected areas, legal status, social license).[10]

~1,000 → 100 → 20 → 10

Funnel figures: illustrative.

Energy Coop candidate community: Corozal, Sucre (IMPE 43.1%) — not a PASO project.[10]

7 Partners and next step

FMO and **RVO** (solar finance/facilitation).[8,9] **Green Choice** and **Energie Samen** — peer learning on energy communities (target; no signed agreements).[10]

Next step: co-design session with executor and candidate community — governance, OPEX, and M&E aligned to IMPE. cleantechhub.net

[6] escuelaing.edu.co — renewables lab and continuing education. [8] fmo.nl, Colombia projects. [9] Solar ecosystem facilitation (MCU 2023). [10] CTH / PASO 2026 (project brief; GRP filter). [11] suncolombia.co — DISPAC project. [12] seosenergy.co. [13] tierragrata.org — clean energy impact.

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